

Лабораторная работа №8

Настройка сетевых сервисов. DHCP

Поляков Г. С.

12 марта 2025

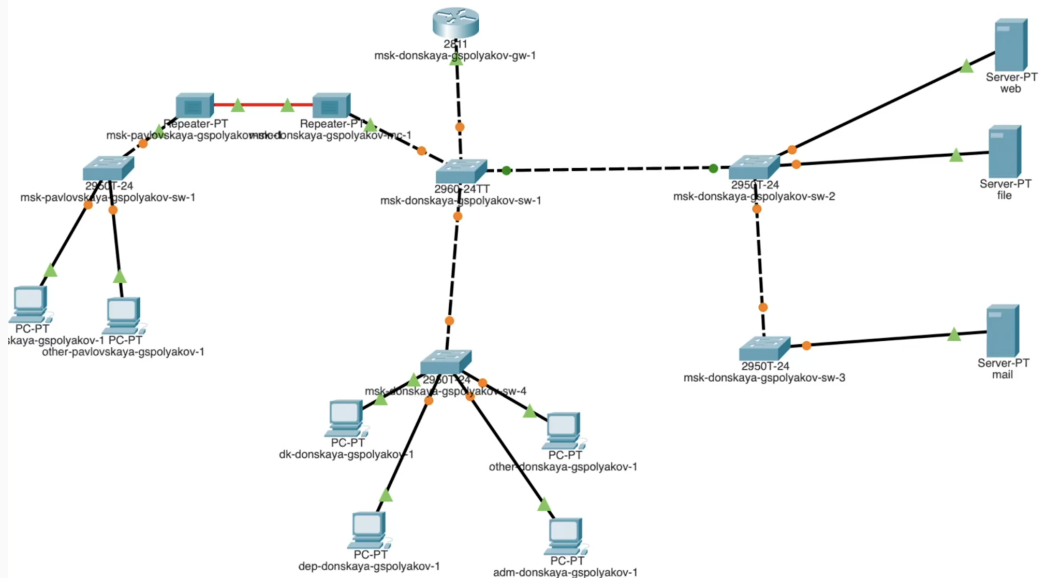
Российский университет дружбы народов, Москва, Россия

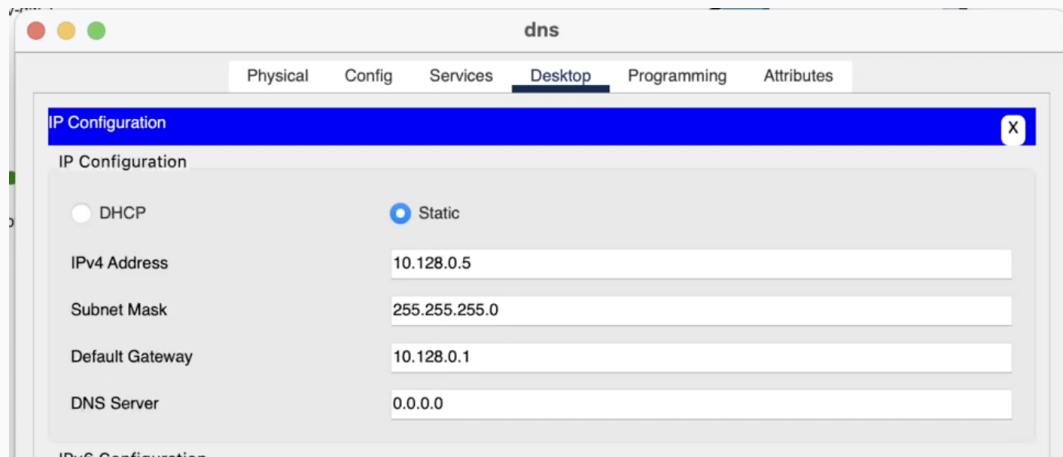
Информация

- Поляков Глеб Сергеевич
- Студент
- Российский университет дружбы народов

Выполнение работы

Схема сети





dns

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service

☒ On

Off

Resource Records

Name

Type

A Record

Address

Add

Save

Remove

No.	Name	Type	Detail
0	dns.donskaya.rudn.ru	A Record	10.128.0.5
1	file.donskaya.rudn.ru	A Record	10.128.0.3
2	mail.donskaya.rudn.ru	A Record	10.128.0.4
3	www.donskaya.rudn.ru	A Record	10.128.0.2

Настройка DHCP

```
msk-donskaya-gspolyakov-gw-1(config)#ip name-server 10.128.0.5
msk-donskaya-gspolyakov-gw-1(config)#ser
msk-donskaya-gspolyakov-gw-1(config)#service dhcp
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp dk
^
% Invalid input detected at '^' marker.

msk-donskaya-gspolyakov-gw-1(config)#ip dhcp pool dk
msk-donskaya-gspolyakov-gw-1(dhcp-config)#net
msk-donskaya-gspolyakov-gw-1(dhcp-config)#network 10.128.3.0 255.255.255.0
msk-donskaya-gspolyakov-gw-1(dhcp-config)#defa
msk-donskaya-gspolyakov-gw-1(dhcp-config)#default-router 10.128.3.1
msk-donskaya-gspolyakov-gw-1(dhcp-config)#dns-s
msk-donskaya-gspolyakov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-gspolyakov-gw-1(dhcp-config)#exi
msk-donskaya-gspolyakov-gw-1(dhcp-config)#exit
msk-donskaya-gspolyakov-gw-1(config)#ip dh
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp ex
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp excluded-address 10.128.3.1 10.128.3.29
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp excluded-address 10.128.3.200 10.128.3.254
msk-donskaya-gspolyakov-gw-1(config)#ip dh
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp po
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp pool de
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp pool departaments
msk-donskaya-gspolyakov-gw-1(dhcp-config)#network 10.128.4.0 255.255.255.0
msk-donskaya-gspolyakov-gw-1(dhcp-config)#default-router 10.128.4.1
msk-donskaya-gspolyakov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-gspolyakov-gw-1(dhcp-config)#exit
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp excluded-address 10.128.4.1 10.128.4.29
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp excluded-address 10.128.4.200 10.128.4.254
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp pool adm
msk-donskaya-gspolyakov-gw-1(dhcp-config)#network 10.128.5.0 255.255.255.0
msk-donskaya-gspolyakov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-gspolyakov-gw-1(dhcp-config)#exit
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp excluded-address 10.128.5.1 10.128.5.29
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp excluded-address 10.128.5.200 10.128.5.254
```



```
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp pool other
msk-donskaya-gspolyakov-gw-1(dhcp-config)#net
msk-donskaya-gspolyakov-gw-1(dhcp-config)#network 10.128.6.0 255.255.255.0
msk-donskaya-gspolyakov-gw-1(dhcp-config)#def
msk-donskaya-gspolyakov-gw-1(dhcp-config)#default-router 10.128.0.5
msk-donskaya-gspolyakov-gw-1(dhcp-config)#default-router 10.128.6.1
msk-donskaya-gspolyakov-gw-1(dhcp-config)#dns
msk-donskaya-gspolyakov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-gspolyakov-gw-1(dhcp-config)#exit
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp excluded-address 10.128.6.1 10.128.6.29
msk-donskaya-gspolyakov-gw-1(config)#ip dhcp excluded-address 10.128.6.200 10.128.6.254
msk-donskaya-gspolyakov-gw-1(config)#exit
```

Информация об привязках выданных устройств

```
msk-donskaya-gspolyakov-gw-l#sh ip dhcp pool

Pool dk :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 0
Excluded addresses : 8
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.128.3.1 10.128.3.1 - 10.128.3.254 0 / 8 / 254

Pool departaments :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 0
Excluded addresses : 8
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.128.4.1 10.128.4.1 - 10.128.4.254 0 / 8 / 254

Pool adm :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 0
Excluded addresses : 8
Pending event : none

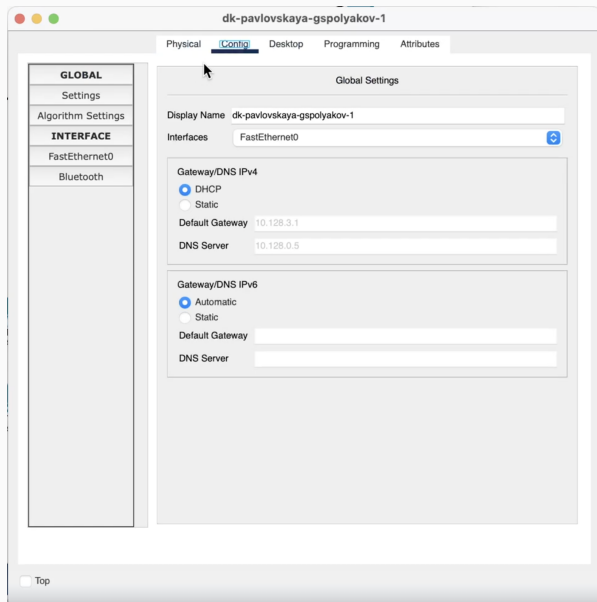
1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.128.5.1 10.128.5.1 - 10.128.5.254 0 / 8 / 254

Pool other :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 0
Excluded addresses : 8
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.128.6.1 10.128.6.1 - 10.128.6.254 0 / 8 / 254
msk-donskaya-gspolyakov-gw-l#
```

```
msk-donskaya-gspolyakov-gw-1#sh ip dhcp binding
IP address      Client-ID/      Lease expiration      Type
                Hardware address
msk-donskaya-gspolyakov-gw-1#
```

Проверка соединения



```
msk-donskaya-gspolyakov-sw-1#ping 10.128.3.1
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 10.128.3.1, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

```
msk-donskaya-gspolyakov-sw-1#
```

